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Interaction of rippled shock wave with flat fast-slow interface

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The evolution of a flat air/sulfur-hexafluoride interface subjected to a rippled shock wave is investigated. Experimentally, the rippled shock wave is produced by diffracting a planar shock wave around solid cylinder(s), and the effects of the cylinder number and the spacing between cylinders on the interface evolution are considered. The flat interface is created by a soap film technique. The post-shock flow and the evolution of the shocked interface are captured by a schlieren technique combined with a high-speed video camera. Numerical simulations are performed to provide more details of flows. The wave patterns of a planar shock wave diffracting around one cylinder or two cylinders are studied. The shock stability problem is analytically discussed, and the effects of the spacing between cylinders on shock stability are highlighted. The relationship between the amplitudes of the rippled shock wave and the shocked interface is determined in the single cylinder case. Subsequently, the interface morphologies and growth rates under different cases are obtained. The results show that the shock-shock interactions caused by multiple cylinders have significant influence on the interface evolution. Finally, a modified impulsive theory is proposed to predict the perturbation growth when multiple solid cylinders are present. *Published by AIP Publishing.* https://doi.org/10.1063/1.5024774

I. INTRODUCTION

The Richtmyer-Meshkov (RM) instability occurs when a shock wave impulsively accelerates a density interface.[1,2] The interface perturbations deposited initially or induced by the rippled shock front grow linearly shortly after the shock compression and then nonlinearly with appearances of spikes and bubbles developed on the interface, and finally the turbulent mixing is induced. Extensive studies have been performed on the RM instability for its significant applications in fields such as inertial confinement fusion (ICF),[3] supersonic combustion,[4] and supernova explosions,[5] and significant advancements have been achieved by experiments, theories, and computations.[6–9]

In the RM instability, the shock and fluid interface provide, respectively, the pressure and density gradients which together contribute baroclinic vorticity to the system if not aligned. Therefore, the shapes of the initial interface and shock are crucial to the perturbation development. In most applications, such as ICF, both the shock and interface are initially disturbed. However, in previous studies on shock-induced interface instability, majority of experiments, numerical simulations, and theoretical analyses dealt with the simple situation, i.e., the initial shock wave is uniform and the perturbations exist only on the initial interface.[10–12] Investigations on the interface evolution caused by perturbed shock waves were rather limited. Ishizaki *et al.*[13] numerically considered the instability of a uniform contact surface impacted by a nonuniform shock wave driven by a rippled piston and found the dependence of the growth rate of the contact interface on the phases of rippled shock waves. Later, the time-variations of amplitudes of the rippled shock wave and contact interface were theoretically compared for a uniform laser irradiation on a perturbed target and for a nonuniform laser irradiation on a smooth target,[14] and approximate formulas for the variations of amplitudes of the shock front and ablation surface were obtained. Generally speaking, a perturbed interface can be produced far more easily than a rippled shock wave in an experiment such that experimental work on this type RM instability is scarce. The interaction between a rippled shock wave and a flat interface with a laser hydrodynamics experiment in a Be shock tube was investigated by Kane *et al.*,[15] in which the rippled shock wave was produced by a planar shock wave passing through a Cu-polyamide interface. Verified by hydrodynamic simulations, the phase and the amplitude of the shocked interface were determined by the velocity perturbation on the shock. Recently, experiments associated with a nonuniform shock were performed by Zou *et al.*[16] in a planar shock tube and by Liang *et al.*[17] in a converging shock tube. The nonuniform shock wave in both cases was produced by diffracting a planar or a cylindrically converging shock wave around one or multiple solid cylinders.

In the work of Zou *et al.*,[16] the amplitude and the distance between two triple points of a rippled shock wave just before impinging the interface were, respectively, considered as the initial amplitude and half of the wavelength of interface to calculate the amplitude growth rate theoretically. The authors reported that the predictions greatly deviate from the experimental counterparts. Note that the membraneless technique was adopted to create the planar interface, and the severe

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gas diffusion will greatly influence the gas concentration near the interface and further the shock movements and the growth rate of the interface amplitude.[18,19] Consequently, the amplitude and wavelength of interface are strongly dependent upon the gas concentration, and the formation of interface which can inhibit the gas diffusion is quite necessary. In addition, the rippled shock wave was produced by diffracting a planar shock wave around one solid cylinder. Therefore, the shock front produced is far from a single-mode shape, and multiple solid cylinders may be helpful for creating a single-mode-like shock front.

In this work, a soap film technique is adopted to form a discontinuous flat interface which suppresses diffusion. A rippled shock wave is produced by a planar shock wave propagating over multiple solid cylinders. The effects of the cylinder number and the spacing of cylinders on the perturbation growth of the flat interface are discussed. Numerical simulations are also performed to acquire more information of the flow field.

II. EXPERIMENTAL AND NUMERICAL METHODS

A. Experimental methods

In this work, the rippled shock wave is produced by diffracting a planar shock around solid cylinders. As shown in Fig. 1(a), two acrylic plates (3 mm in thickness) are first manufactured by engraving grooves with a diameter of 20.1 mm and a depth of 1 mm, and then both ends of each solid cylinder with a height of 22 mm and a diameter (d) of 20 mm are, respectively, inserted into the grooves on the two Acrylic plates such that the separation of the two plates is 20 mm. A flat interface is created by the soap film technique, which has already verified its feasibility and reliability in our previous work.[20,21] Unlike previous studies, no wires are needed in this work. Before creating interfaces, the framework edges should be wetted by a solution made of 78% distilled water, 2% sodium oleate, and 20% glycerine (the numbers denote percent by mass). Afterwards, a rectangular frame with the soap film on its surface is pulled along the interface framework, and a flat interface can be created. Subsequently, the device is inserted into the test section and fitted tightly with optical windows on top and bottom sides of the test section. In order to generate an air/sulfur-hexafluoride (SF_6) interface, air at the right side of the interface should be removed and replaced by SF_6 prior to the experiment. For this purpose, SF_6 is injected into the test section through an "Inflow" hole, and air is gradually exhausted through an "Outflow" hole, as shown in Fig. 1(b). During this process, an appropriate inflating ratio is required to protect the formed interface. A gas concentration detector is placed at the "Outflow" hole to monitor the purity of SF_6 at the right side of the interface. In the experiment, when the concentration of oxygen is lower than 1% by volume fraction, it is considered that air at the right side of the interface has been replaced by SF_6 completely. The experiment is conducted within 20 s to reduce the contamination of SF_6 by air.

The experiments are preformed in a horizontal shock tube, which consists of a 1.7 m long driver section, a 3.9 m long driven section, and a 1.0 m long test section. The rectangular cross section of the test section is 155 mm × 26 mm. The Mach number of the incident planar shock wave (M_s) is maintained at 1.21 ± 0.01. Illuminated by a DC regulated light source (CEL-HXF300, the maximum power output is 249 W), schlieren photography combined with a high-speed video camera (FASTCAM SA5, Photron Limited), as shown in Fig. 1(b), is used to visualize the flow field. The frame rate of the high-speed video camera is 50 000 fps (corresponding to a time interval of 20 μs), and the exposure time is 0.366 μs for all cases. In this work, five different cases are involved, considering the effects of the solid cylinder number and spacing of solid cylinders on interface evolution. Due to the greater activity of air than SF_6, the gas at the right side of the interface is actually a mixture of SF_6 and air. The composition of the mixture calculated by one-dimensional (1D) gas dynamics and the other initial experimental parameters are listed in Table I. l is the distance of the initial flat interface from the cylinder center, which is 20 mm for all cases. l_r is the spacing of cylinder centers, and a dimensionless distance ϵ is defined as $\epsilon = l_r/d$.

B. Numerical method

The multi-component flows discussed in this study are modeled using the compressible Euler equations augmented by one species equation for the interface gas (SF_6). For the time-dependent two-dimensional case, the governing equation can be written in the form

$$\frac{\partial \boldsymbol{U}}{\partial t} + \frac{\partial \boldsymbol{F(U)}}{\partial x} + \frac{\partial \boldsymbol{G(U)}}{\partial y} = 0, \qquad (1)$$

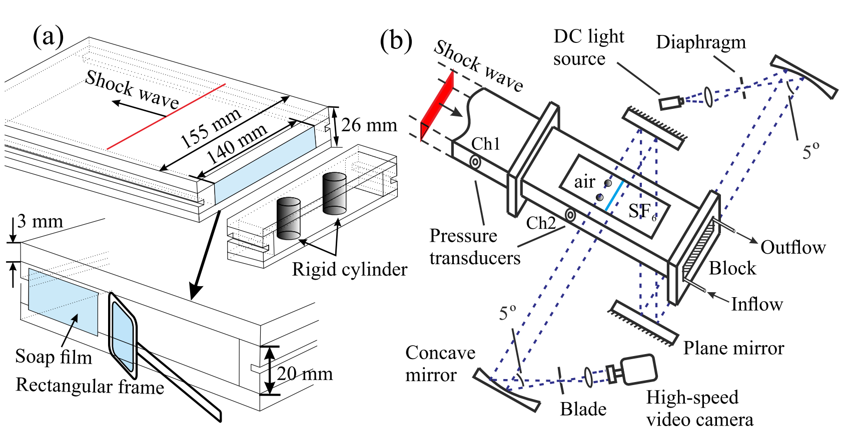

FIG. 1. (a) Schematics of relative positions of solid cylinders to the interface (top) and the device to generate the soap film interface (bottom). (b) The shock tube test section with the schlieren system.

TABLE I. Experimental initial parameters for all cases. The numbers behind the letters C and E represent the number of cylinders and the proximate ratio of the spacing between cylinders with the diameter of cylinder, respectively; n is the number of solid cylinder; l_r is the spacing of the centers of solid cylinders; $\epsilon = l_r/d$ with d being the cylinder diameter; MF means the mass fraction of SF_6 in the test gas. A^+ is the postshock Atwood number; Δv is the jump velocity of interface impacted by the incident shock wave; "PR" means the pixel resolution.

Case	n	ϵ	l_r (mm)	MF (%)	A^+	Δv (m/s)	PR (mm/pixel)
C1E0	1	0	0	93.0	0.58	76.21	0.29
C2E2	2	2.0	40.0	92.5	0.58	76.59	0.30
C3E2	3	2.0	40.0	92.2	0.59	76.68	0.30
C2E1	2	1.33	26.67	89.4	0.55	78.53	0.30
C2E3	2	3.0	60.0	92.0	0.59	76.76	0.26

where $\boldsymbol{U}$ is the conserved variables, $\boldsymbol{F}(\boldsymbol{U})$ and $\boldsymbol{G}(\boldsymbol{U})$ represent the convective fluxes in the x- and y-directions, respectively,

$$\boldsymbol{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \\ \rho Y_s \end{pmatrix}, \boldsymbol{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho u v \\ (\rho E + p)u \\ \rho Y_s u \end{pmatrix}, \boldsymbol{G} = \begin{pmatrix} \rho v \\ \rho v u \\ \rho v^2 + p \\ (\rho E + p)v \\ \rho Y_s v \end{pmatrix}, \quad (2)$$

p, ρ, E, u, and v, respectively, represent the pressure, density, total energy per unit mass, and velocity components in the x- and y-directions of the mixture. Y_s represents the mass fraction of gas at one side of the interface, and the mass fraction of gas a at the other side of the interface is $Y_a = 1 - Y_s$. The equation of state of the mixture is expressed as $p = \rho T(Y_s R_s + Y_a R_a)$, where R_s and R_a represent the gas constants of gases s and a and T is the temperature of the mixture. E is the total energy of the mixture and $E = Y_s e_s + Y_a e_a + (u^2 + v^2)/2$, where e_s and e_a are the internal energies of gases s and a.

The finite volume method is used to discretize the conservation laws. The MUSCL (monotonic upstream-centered scheme for conservation laws)-Hancock scheme is adopted to achieve the second-order accuracy in both temporal and spatial scales. The HLL (Harten-Lax-van Leer) Riemann solver is employed to evaluate the physical fluxes through the cell interfaces. The unstructured adaptive mesh is employed so that it refines local complex areas and could effectively capture wave patterns including shocks, expansion waves, and interfaces in compressible flows, such as shock-body interactions[22] and shock-interface interactions.[23]

As shown in Fig. 2(a), due to the symmetric structure of the whole physical space, the upper-half plane ($0 \le x \le 150$ mm and $0 \le y \le 70$ mm) is considered as the computational domain. The left and right boundaries ($x = 0$ and $x = 150$ mm) are treated as inflow and outflow conditions, respectively, while the upper and lower boundaries ($y = 0$ and $y = 70$ mm) are considered as reflection and symmetry conditions, respectively. In order to validate the mesh convergence, a planar shock wave diffracting around a solid cylinder is considered, in which a uniform mesh without grid adaptation is employed with the initial mesh size of 0.8, 0.4, 0.2, and 0.1 mm, respectively. The pressure profiles along the horizontal symmetry axis of the flow field with different initial mesh sizes are given in Fig. 2(b). It can be found that the values are convergent when the mesh sizes come to 0.2 mm or 0.1 mm. Therefore, in order to ensure the accuracy and meanwhile to minimize the computation capacity, the initial mesh size of 0.2 mm is adopted in computations and 0.025 mm is used where greater density gradient exists.

III. WAVE PATTERNS

Figure 3 presents typical schlieren images of wave patterns after a planar shock wave diffracts around a solid cylinder for the C1E0 case and around two solid cylinders for the C2E2 case without a gas interface. In the C1E0 case, when an incident shock wave (IS) meets the cylinder, a regular reflection occurs accompanied by a reflected shock (RS). As the angle of shock front with the cylinder surface increases, a Mach reflection with a Mach stem (MS1) occurs, as presented in Fig. 3(a). Subsequently, two Mach stems (MS1) at both sides of the cylinder meet each other at the eastern pole, resulting in an additional Mach stem (MS2) propagating forwards and diffracted shock waves (DS) traveling along the cylinder periphery [Figs. 3(b) and 3(c)]. As time proceeds, both MS1 and MS2 increase in length at the expense of length reduction of the uniform IS, as indicated in Fig. 3(d). For the C2E2 case, as the IS moves along the cylinder surfaces, a pair of RS collides with each other [Fig. 3(e)]. Afterwards, interactions between the pair of RS and the uniform IS lead to the formation of a new Mach stem (MS3), as indicated in Fig. 3(f). With time going on, the

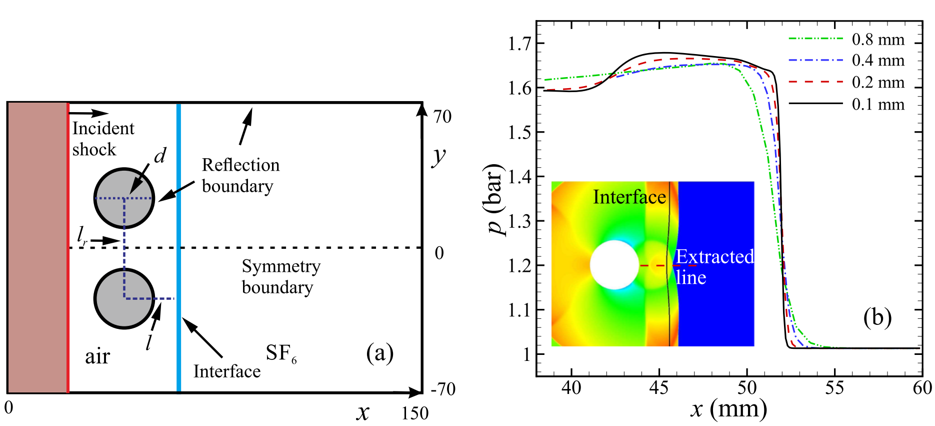

FIG. 2. Initial configurations of numerical simulation (a) and grid convergence validation in the C1E0 case (b).

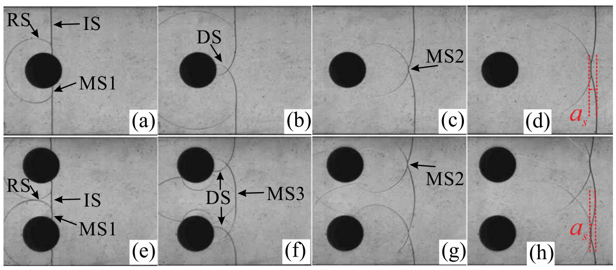

FIG. 3. Schlieren frames of wave patterns after a planar shock impacts one solid cylinder (upper) and two solid cylinders (lower) at four moments. IS, incident shock wave; RS, reflected shock wave; DS, diffracted shock wave; MS, Mach stem; a_s, amplitude of disturbed shock wave.

length of MS3 decreases while the length of MS2 increases, and the curvature of MS3 gradually reduces [Figs. 3(g) and 3(h)].

The Mach number and amplitude of rippled shock wave when interacting with a flat interface are crucial to the interface development. The Mach number of Mach stems is calculated by $Ma = v/c$ with v and c being the average velocity of Mach stem and local sound speed, respectively. Due to the nonuniform feature of the Mach stem, the average velocity v is used by $v = (x_{i+2} - x_i)/2\Delta t$, in which x_i is the position of the Mach stem at time i and x_{i+2} is the position of the Mach stem at time $i + 2\Delta t$ with Δt being the time internal between two neighboring frames. Time-variations of the Mach number of the Mach stem are shown in Fig. 4(a). The dimensionless time tv_s/d is adopted where v_s is the velocity of the IS, and the initial time is defined as the moment of the IS passing across the cylinder center. In general, the strength of all Mach stems decreases with time, and the reduction rates of MS2 in the C1E0 case and MS3 in the C2E2 case are greater relatively. Particularly, the evolution of MS3 in the C2E2 case exhibits a "break," i.e., the slope changes, at late time. Note that the occurrence of the "break" is due to the collision of MS3 with the intersection of the diffracted shock waves, enhancing the intensity of MS3. Meanwhile, the amplitudes of rippled shock wave a_s, as defined in Figs. 3(d) and 3(h), are presented in Fig. 4(b). The amplitudes decrease sharply for both cases at first and then slowly in the C1E0 case but still greatly in the C2E2 case due to the great change of MS3. The continuous and great reduction of the amplitude in the C2E2 case even results in a negative amplitude, i.e., a phase inversion of the shock front. Overall, the coupling of multiple solid cylinders leads to a diverse evolution of the rippled shock wave compared with one solid cylinder case and further to diverse morphologies of the interface.

For a sinusoidal-shaped shock wave, the shock stability problem as an initial-value problem was theoretically investigated by Erpenbeck,[24] in which the mathematical approach avoided the derivation of an integral equation in the time domain but required a complicated, inverse Laplace-transform operation to ascertain the temporal evolution of disturbances at the shock's surface. Bates[25] also derived linear instability criteria for an isolated, planar, two-dimensional shock wave propagating through an inviscid fluid with an arbitrary equation of state. The basis for this analysis was a novel solution for the time-dependent Fourier amplitude of a single-mode perturbation on the front, which was expressed in the form of a Volterra equation. For these two disparate methodologies, Swan and Fowles[26] and Tumin[27] confirmed that the results derived by Erpenbeck are exactly the same as the results derived by Bates. In this work, the theory proposed by Bates is adopted to predict the shock behaviors for the C2E2 case. In the theory, the initial sinusoidal-shaped shock wave is assumed and the amplitude and the wavelength are the same as those in the experiment (the wavelength is the distance between centers of two cylinders). The postshock flow parameters before the shock encounters the cylinder are adopted. The analytical prediction is given in Fig. 4(b) by a solid line, which is roughly consistent with the experimental result. Note that the shape of the rippled shock wave in the experiment is not exactly sinusoidal. Also after the shock diffracts around the cylinders, the postshock flow is unsteady and additional high-pressures will be induced by shock-shock interactions. The complicated pressure distributions in the postshock flow result in the differences between experimental and analytical results.

Note that the phase inversion of the shock front is directly associated with the position of the "break." Consequently, it would be interesting to relate the position of the "break" to the length scales in the problem, like the radius of the cylinders and the separation between them. For this purpose, we numerically investigate the evolution of MS3 for two cylinder case, considering the effect of a dimensionless parameter $\varepsilon = l_r/d$. Five different ε are involved, and the variations of the Mach number of MS3 are given in Fig. 5(a). The results

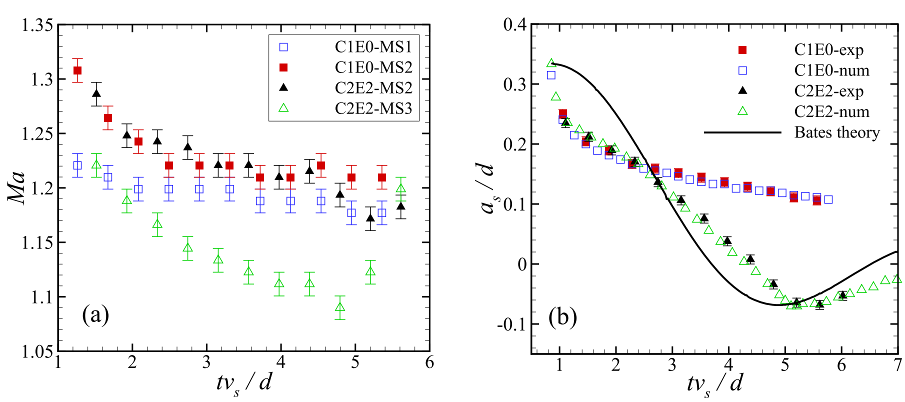

FIG. 4. Variations of the Mach number of the Mach stem (a) and variations of the amplitude of rippled shock wave (b).

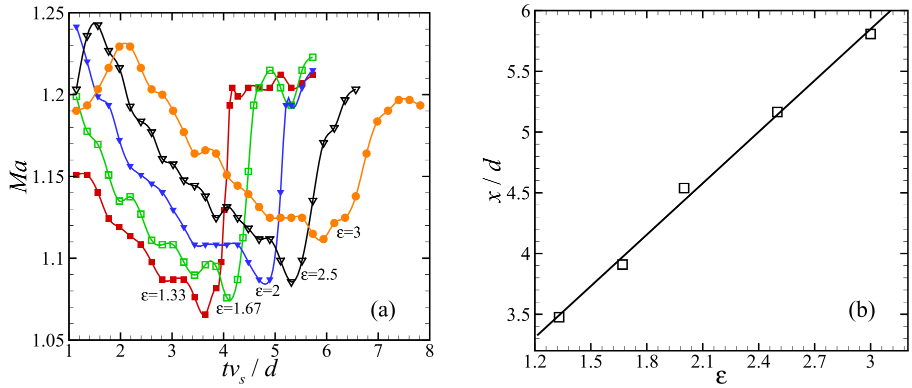

FIG. 5. Time-variations of the Mach number of MS3 in two cylinder cases (a) and variations of dimensionless positions of the "break" of MS3 with dimensionless parameter ε (b).

show that as ε increases, the moment of the occurrence of the "break" is delayed. Further, the position of the "break" varies as ε changes, and their relationship is presented in Fig. 5(b), in which x is the distance between the position of the "break" and the cylinder center. A better linearity is observed, and the result indicates that as ε increases, the position of phase inversion of the shock front is postponed. This result is significant for determining the position of the initial interface.

IV. INTERFACE MORPHOLOGY

A. Single solid cylinder

Schlieren images of a flat air/SF_6 interface impacted by a rippled shock wave for the C1E0 case from the experiment and numerical simulation are shown in Fig. 6. The initial time is defined as the moment when the planar IS meets the interface. In this work, early evolution of wave patterns and interface evolution can be captured compared with previous work,[16] which is beneficial for us to determine the "initial" interface amplitude induced by the rippled shock wave. Due to the limited distance of the solid cylinder from the interface, before the formation of MS2, a chevron-like rippled shock wave encounters the flat interface, generating a transmitted shock (TIS) and a backward reflected shock (BRS) (t = 69.78 μs). The interface is imprinted by a chevron-like perturbation and, as time goes on, the perturbation amplitude increases (t = 69.78-969.78 μs). During this process, no obvious vortices are generated because of weak baroclinic effect. It should be noted that the soap film technique shares the same drawback with other membrane techniques for generating interface, i.e., the membrane will absorb some energy from the shock impact and may have some possible effects on the onset of the RM instability. However, as Cohen[28] pointed out that the effects of the soap films were negligible because the thickness of the soap films is very thin. As also discovered in our results, good agreement of the interface morphologies is obtained from the comparison between the experimental and numerical results. Therefore, it is believed that the soap films have limited effects on the onset of the RM instability.

The variation of the interface amplitude, indicated in Fig. 6, is shown in Fig. 7. Generally, the interface amplitude grows almost linearly. At some moments, as indicated by dashed rectangles, the amplitude growth is ceased, which is caused by reflection shock waves from the solid cylinder and tube walls. The experimental results coincide with the numerical counterparts, and the growth rates of the amplitude are ~5.56 m/s and 5.53 m/s, respectively. To theoretically investigate the growth rate of the amplitude, the initial amplitude and wavelength of the perturbed interface must be determined in advance. The previous work[16,17] showed that the interface perturbations produced are notably smaller than the disturbances

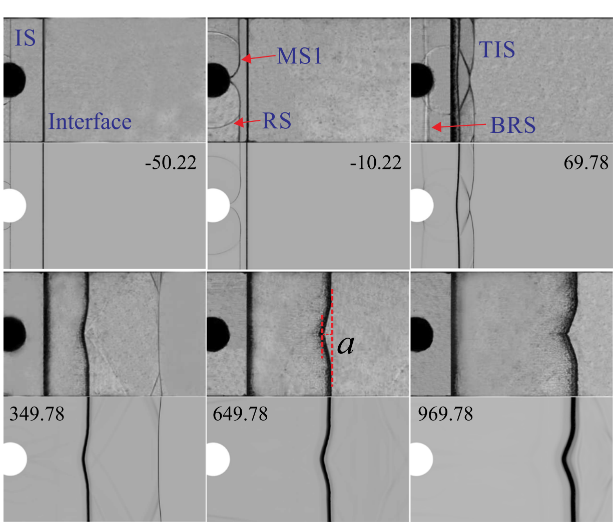

FIG. 6. Evolution of a flat air/SF_6 interface impacted by a rippled shock wave for the C1E0 case.

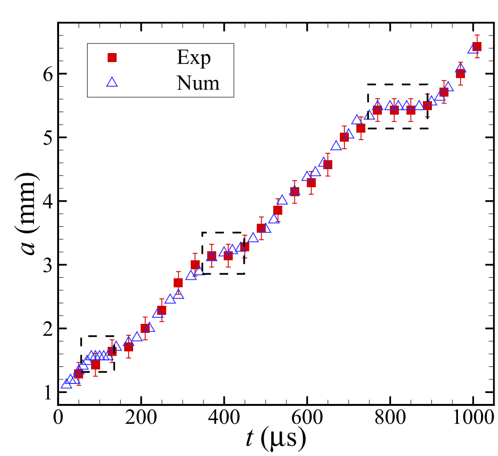

FIG. 7. Time-variation of the distorted interface amplitude for the C1E0 case from the experiment and numerical simulation.

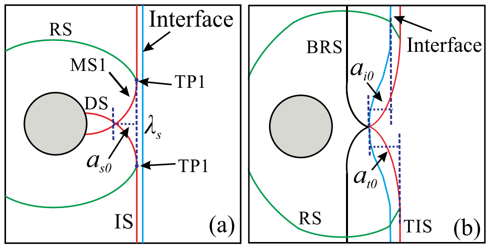

FIG. 8. Sketches of wave patterns shortly before (a) and after (b) the rippled IS impacts the interface, respectively.

causing them, and the perturbation on the rippled shock wave (a_{s0}) cannot be equivalent to the perturbation on the interface (a_{i0}), as indicated in Fig. 8. Liang *et al.*[17] proposed to adopt the compressed interface amplitude in the experiment as an initial one and concluded that better agreement between theoretical and experimental results can be achieved.

Note that a_{i0} is closely dependent on the initial planar shock strength M_s, the diameter of solid cylinder d, and the distance of cylinder from the interface l. For figuring out the relationship among them, extensive numerical simulations on the interaction of a rippled shock wave with an air/SF_6 planar interface are performed, considering different initial planar shock strengths and different cylinder diameters. The time-variations of the perturbation on the rippled IS (a_s) and the strength of MS2 (Ma) are given in Figs. 9(a) and 9(b), respectively, and the characteristic quantities are $M_s d$ and $M_s^{0.1}$, respectively. Both dimensionless perturbation and strength decrease almost linearly before the dimensionless time of 1.0 and then change nonlinearly with different laws. According to their nonlinear behaviors, a power function is adopted for the dimensionless perturbation and a rational function is adopted for the dimensionless shock strength in the nonlinear stage. Combined with the linear stage, the variation of the dimensionless perturbation on the rippled IS can be described as

$$f_1(\tau) = \begin{cases} -0.298\tau + 0.506 & (0.88 \le \tau \le 1.0) \\ (21.57\tau)^{-0.56} + 0.027 & (\tau > 1.0) \end{cases}, \tag{3}$$

and the variation of the dimensionless strength of MS2 can be written as

$$f_2(\tau) = (1.089 - 1.356\tau)/(0.9436 - 1.131\tau)(\tau \ge 0.88). \tag{4}$$

When the uniform IS meets the planar interface, the perturbation amplitude of the rippled IS can be calculated by Eq. (3), i.e., $a_{s0} = f_1(\tau_1)$, where $\tau_1 = l/d$. Suppose that after a dimensionless time τ_2, the rippled IS completely passes through the planar interface, then the following expression can be established:

$$M_s^{-1.9} \int_{\tau_1}^{\tau_2} f_2(\tau) d\tau = f_1(\tau_1), \tag{5}$$

from which the dimensionless time τ_2 can be obtained. Subsequently, the initial perturbation of the TIS (a_{t0}) can be obtained through the following expression:

$$a_{t0} = (\tau_2 - \tau_1) v_t d / v_s, \tag{6}$$

where v_t is the velocity of undisturbed transmitted shock wave and can be calculated by 1D gas dynamics theory.

In the previous work,[1,29] a simple kinematic relationship was proposed to show the amplitudes of the TIS and the RS after a planar shock wave impacts a sinusoidal fast-slow interface. In the present study, however, the perturbation directions of the TIS and the RS are opposite, which is different from the

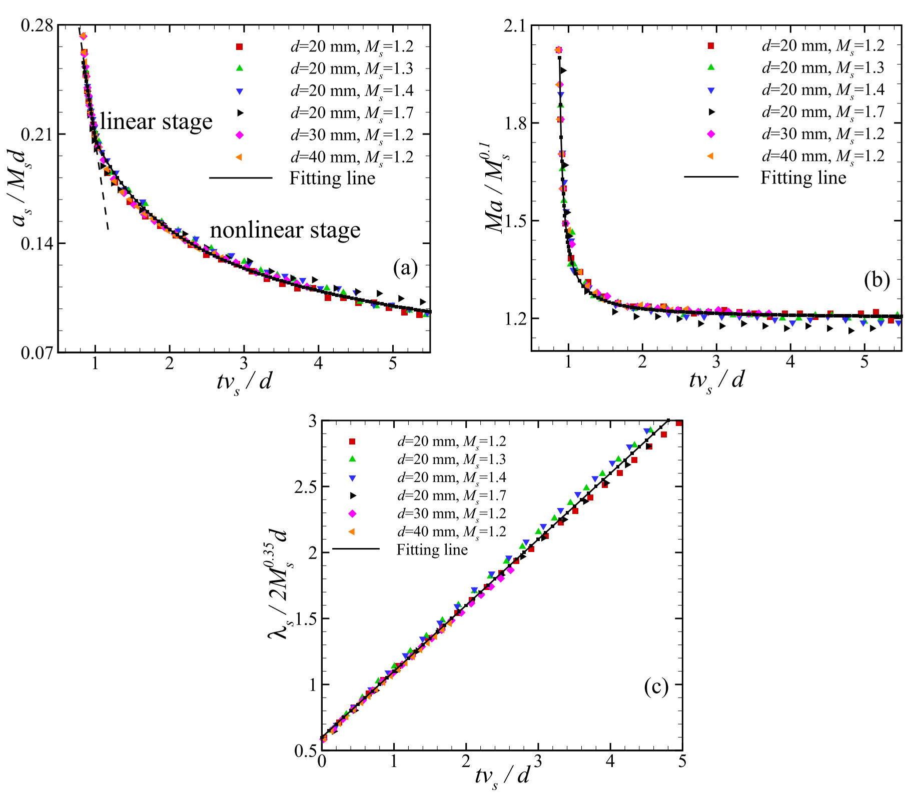

FIG. 9. Time-variations of the perturbation (a), Mach number (b), and wavelength (c) of the rippled IS.

TABLE II. The characteristic parameters and comparison between numerical and theoretical amplitudes under different shock Mach numbers and different cylinder diameters.

M_s	D (mm)	v_s (m/s)	v_t (m/s)	v_r (m/s)	a_{tnum} (mm)	a_{t0} (mm)	a_{inum} (mm)	a_{i0} (mm)
1.2	20	418.6	178.6	286.3	1.95	1.84	0.87	0.88
1.3	20	443.8	200.0	259.3	2.02	2.09	1.15	1.04
1.4	20	477.9	221.7	234.5	2.48	2.65	1.47	1.36
1.7	20	580.3	287.9	169.3	3.10	4.16	2.34	2.33
1.2	30	418.6	178.6	286.3	2.97	2.77	1.35	1.31
1.2	40	418.6	178.6	286.3	3.95	3.69	1.68	1.75

previous work.[1,29] Here a new relationship between the perturbation of the TIS and the initial disturbed amplitude (a_{i0}) of the interface is proposed,

$$a_{i0} = \frac{a_{t0}}{1 + \frac{v_t + v_r}{v_s}} \tag{7}$$

with v_r being the RS velocity which can be obtained through 1D gas dynamics theory.

To verify the relationship in Eq. (7), numerical simulations considering different incident shock Mach numbers and different diameters of the solid cylinder are performed, and the initial perturbations of the TIS (a_{tnum}) and the deformed interface (a_{inum}) are measured, as listed in Table II. The corresponding values calculated by Eqs. (6) and (7) (a_{t0} and a_{i0}) are also given in Table II. It can be found that the initial amplitudes of the deformed interface calculated from the theory agree well with the numerical ones, especially for weak shock wave conditions.

After the interface amplitude is known, the interface wavelength will be determined. In the previous work,[16,17] the distance between two triple points is considered as one half of the wavelength. As a result, it is necessary to determine the locus of the triple point. In fact, it is of fundamental interest to investigate the locus of TP1 when the IS diffracts around a solid cylinder by experiments,[30,31] numerical simulations,[32–34] and theories.[35] After the IS passes across the center of the cylinder, a nearly straight locus of TP1 is found, as shown in Fig. 9(c). Through the linear fitting, the wavelength of rippled shock wave λ_s can be written as

$$\lambda_s/2 = (0.5\tau + 0.6)M_s^{0.35}d. \tag{8}$$

To verify Eq. (8), the amplitude growth rate in the C1E0 case will be calculated. Based on Eqs. (7) and (8), the compressed interface amplitude (a_{i0} = 0.88 mm) and the wavelength of rippled shock wave (λ_s = 43 mm) can be obtained. The calculation by the impulsive model[1] is justified because the ratio of $a_{i0}/\lambda_s < 0.1$ is sufficiently small,

$$v_{the} = a_{i0}\frac{2\pi}{\lambda}A^+\Delta v, \tag{9}$$

and the growth rate of the amplitude is calculated to be 5.53 m/s, which is consistent with the experimental and numerical ones, verifying the relationship between the rippled shock wave and compressed interface mentioned earlier. Moreover, the experimental results in the work of Zou *et al.*[16] are compared with the theoretical predictions, as shown in Table III, where λ_s is the wavelength of interface calculated by Eq. (8) and $\lambda_{\exp}$ is the wavelength of rippled shock wave impacting the interface obtained from experiments. Note that the wavelength defined in this work is half of that in the previous work.[16] λ_s is larger than $\lambda_{\exp}$ in three cases, which is ascribed to the pollution of nitrogen by SF_6, resulting in the slower movement of triple points, as discussed before. As a result, the theoretical growth rate of the amplitude calculated with λ_s (v_{the}) is smaller than that calculated with $\lambda_{\exp}$ (v'_{the}). However, good agreement is achieved between the experimental results (v_{exp}) and v'_{the}, further verifying the method proposed.

In this section, we also numerically investigate the effect of the initial planar shock Mach number on the interface amplitude variation for a single solid cylinder. The variations of the dimensionless amplitude for four cases are given in Fig. 10. As the shock wave strengthens, the initial perturbation amplitude of the interface obtained is larger, which can also be observed in Table II. Also the amplitude growth rate is in proportion to the shock intensity. Besides, as the shock Mach number increases, the startup process is extended. In other words, more time is needed for the amplitude growth rate reaching a steady state when the shock wave is stronger. A similar conclusion was reached by Lombardini and Pullin[36] in the classical RM instability problems. Actually, as the shock Mach number varies, the pressure perturbations caused by shock-shock interactions, which are directly associated with the interface evolution, will

TABLE III. Comparison of amplitude growth rates in previous work[16] with theoretical predictions introduced in this study.

Case	M_s	D (mm)	l/d	$\lambda_{\exp}$ (mm)	λ_s (mm)	$v_{\exp}$ (m/s)	v_{the} (m/s)	v'_{the} (m/s)
1	1.22	10	2	26.4	34.3	4.68	3.55	4.61
2	1.21	6	3.3	19.7	22.5	2.94	2.45	2.81
3	1.23	10	4	39.8	55.9	2.25	1.70	2.38

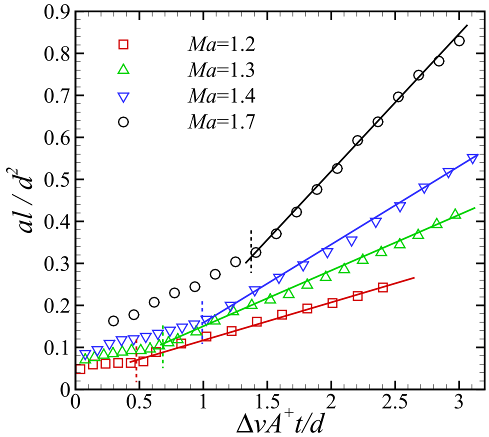

FIG. 10. Time-variations of the distorted interface amplitudes for the single cylinder case with different shock Mach numbers. The vertical dashed lines represent the moments of the end of the startup process and the beginning of the linear stage.

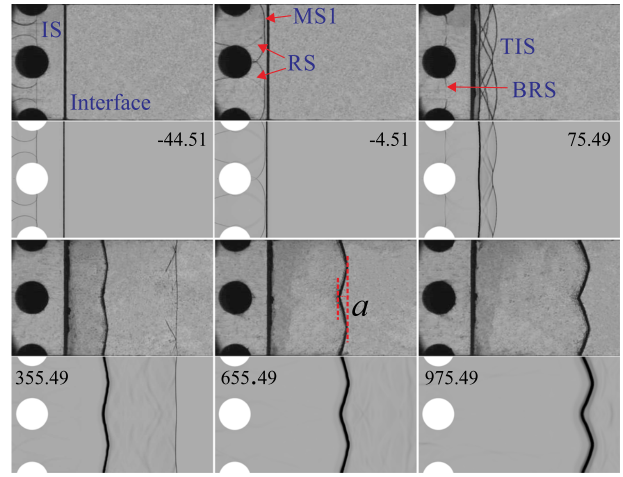

FIG. 12. Evolution of a flat air/SF_6 interface impacted by a rippled shock wave for the C3E2 case.

behave quite differently. Particularly, when the shock intensity is large enough, the compressibility in the postshock flow cannot be ignored, and more complicated flow features may be exhibited.

B. Effects of number of solid cylinders

Figures 11 and 12 present schlieren images of interface evolution for the C2E2 and C3E2 cases, respectively, considering the effects of the cylinder number on perturbation growth. Note that the border of the flow field is 30 mm away from the tube wall for the two cases. Due to the small distance between the cylinder and the interface, the single-mode shock front is not formed absolutely because multiple shock waves are present. In the C2E2 case, after the rippled shock wave impacts the initial flat interface, two chevron-like interfaces are generated. It is believed that if the distance between cylinders is long enough, the interface will behave similar to the C1E0 case. In the C2E2 case, however, due to the appropriate distance between two cylinders, a pair of the RS interacts with each other at a point on the IS before the planar IS impacts the flat interface. Then, MS3 with a curved shape grows and is among the first to impact the flat interface, resulting in a direct connection of the two chevron-like interfaces. In the C3E2 case, similarly, three chevron-like interfaces are generated and connected directly. Overall, the interaction between a planar shock wave with multiple solid cylinders results in complicated wave patterns, which have effects on interface morphologies.

Amplitude variations of the distorted interface are compared in Fig. 13 for cases with different cylinder numbers. The temporal and spatial scales are characterized by $d/\Delta v A^+$ and d^2/l, respectively. At early stages, the interface amplitudes in the C2E2 and C3E2 cases decrease with time. Compared with the C1E0 case, the RS in the C2E2 case generated from one solid cylinder will impact the chevron-like interface near the other solid cylinder. The high pressures behind the RS accelerate the motion of the chevron-like interface and reduce the growth rate of the amplitude. In the C3E2 case, the interface behaviors near the side cylinders are analogous to the C2E2 case. However, the RS from the side cylinders will interact with each other at the center, as shown in Fig. 12, and the induced high pressures will speed up the central interface

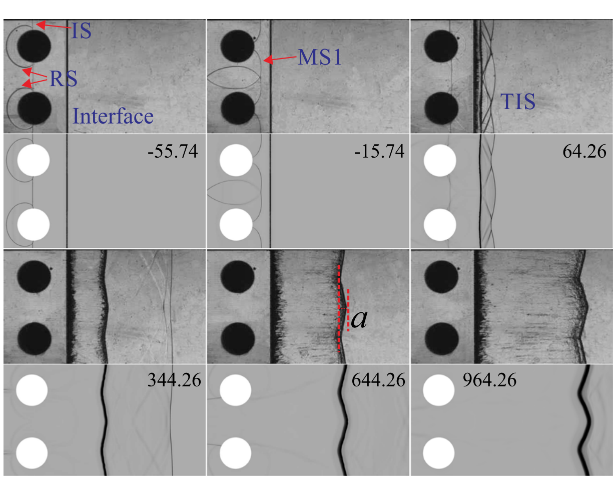

FIG. 11. Evolution of a flat air/SF_6 interface impacted by a rippled shock wave for the C2E2 case.

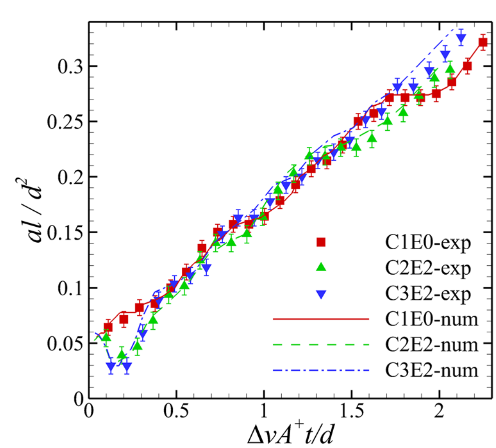

FIG. 13. Time-variations of the distorted interface amplitudes for C1E0, C2E2, and C3E2 cases.

TABLE IV. Comparison of amplitude growth rates in four cases obtained from the experiment, numerical simulation, and theoretical model.

Case	a'_{i0} (mm)	λ'_s (mm)	v_{exp} (m/s)	v_{num} (m/s)	v_{the} (m/s)
C2E2	0.76	40.0	5.46	5.62	5.52
C3E2	0.58	40.0	6.15	6.22	6.17
C2E1	0.15	26.67	6.57	6.67	2.29
C2E3	1.06	43.0	4.64	4.85	4.67

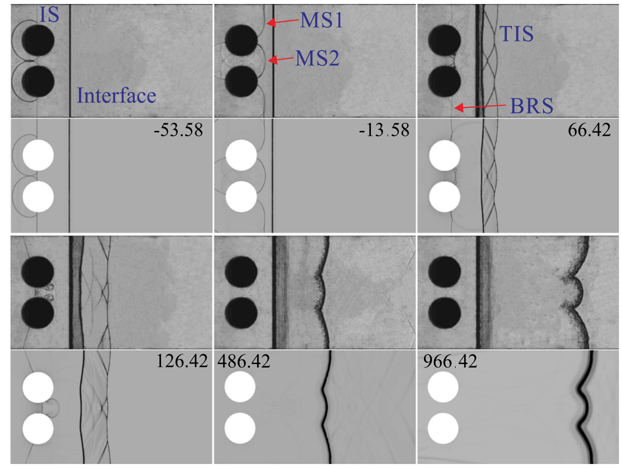

FIG. 15. Evolution of a flat air/SF_6 interface impacted by a rippled shock wave for the C2E1 case.

movement. As a result, the interface amplitude at the center will reduce more heavily. Subsequently, the amplitude grows quickly because high pressures are generated at the right side of the chevron-like interface due to the interaction of the TIS, which inhibits the motion of the chevron-like interface and promotes the interface amplitude.

The impulsive theory can be modified to predict the growth rate of the amplitude under different solid cylinder cases,

$$v_{the} = a'_{i0}\frac{2\pi}{\lambda'_s}A^+\Delta v\frac{n}{\epsilon}, \tag{10}$$

where a'_{i0} is the amplitude after the RS compression and $\lambda'_s = \min(l_r, l_{TP1})$ with l_{TP1} being the spacing of the two TP1. The linear growth rates from experiments, computations, and theoretical predictions in the C2E2 and C3E2 cases are provided in Table IV, and good agreement among them is reached. The growth rate in the C3E2 case is larger than that in the C2E2 case, which indicates that complicated shock-shock interactions among multiple solid cylinders have significant influence on the interface evolution.

C. Effects of distance between solid cylinders

Figures 14 and 15 present schlieren images of interface evolution for the C2E3 and C2E1 cases, respectively, considering the effects of the distance between two cylinders (l_r). In the C2E3 case, due to a large l_r, two chevron-like interfaces are connected by a nearly planar interface because the RS pair has not interacted with each other yet when the planar shock wave encounters the planar interface. In the C2E1 case, however, the connection between two chevron-like interfaces is an arc-shaped one, which is directly associated with the shape of MS3, as indicated in Fig. 15 at $t = -13.58$ μs. The arc-shaped interface moves relatively slower than the interfaces on the side cylinders because the strength of MS3 is weaker than that of MS1. In this case, it seems that the interface consists of bubbles with different wavelengths and different amplitudes, which differs from the C2E2 case where the interface can be decomposed into bubbles with the same amplitude and wavelength. Therefore, the bubble competition may occur in the C2E1 case at late stages, which is similar to previous work.[37] Moreover, as shown at $t = 126.42$ μs in the C2E1 case, a vortex pair is observed between two solid cylinders from both the experimental and numerical results. Such a vortex pair has also been observed in the previous studies[33,38,39] but has not been explained definitely. For illustrating the formation mechanism of the vortex pair, a detailed numerical simulation has been performed, as shown in Fig. 16. During the diffraction of the IS around the solid cylinder, the strength of MS1

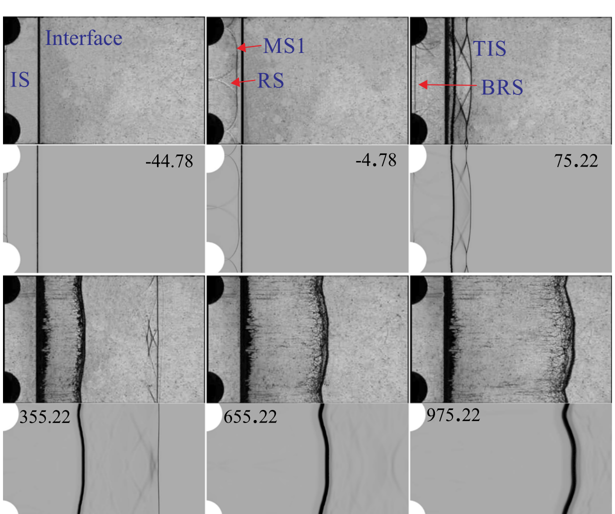

FIG. 14. Evolution of a flat air/SF_6 interface impacted by a rippled shock wave for the C2E3 case.

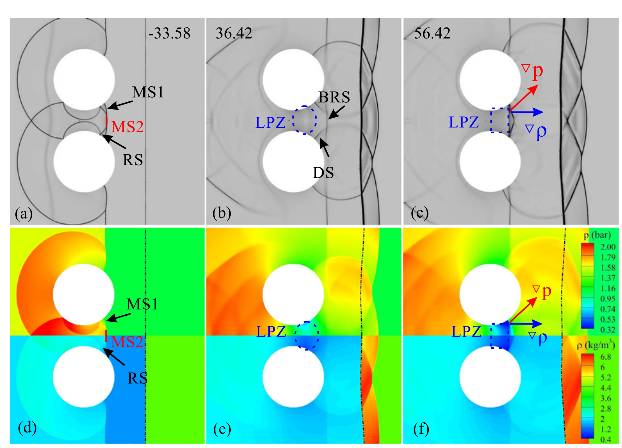

FIG. 16. Schlieren frames of wave patterns in the C2E1 case [(a)-(c)] and the corresponding pressure contours (upper) and density contours (lower) [(d)-(f)], respectively.

decreases, resulting in a low pressure zone (LPZ) near the downstream interface surface. Because of the baroclinic mechanism, the vortex pair is generated near the interface surface. In the limit where the cylinders are touching, the net effect should be similar to a single cylinder with twice the radius to a large extent. However, we can imagine that the pressure perturbations caused by wave patterns at the downstream center would show some differences, and the induced interface morphologies will be imprinted by these differences.

The time-variation of the distorted interface amplitude for cases with different distances between cylinders is shown in Fig. 17. After the rippled shock wave passes across the flat interface, the amplitudes of the shocked interface in the C2E1 and C2E2 cases first decrease because of the compression of the RS and then grow gradually. In the C2E3 case, however, the interface amplitude initially grows, then decreases, and finally increases again. Due to the longest distance between cylinders, the amplitude grows at first with a similar way to the C1E0 case. However, the RS reflected from solid cylinders will still compress the interface, causing a reduction of the interface amplitude. Due to the weakest RS in the C2E3 case, the amplitude reduces less heavily. Relatively, the RS strength in the C2E1 case is the largest, leading to a greater reduction of amplitude. The modified impulsive theory [Eq. (10)] can also be adopted to predict the amplitude growth rate in the C2E1 and C2E3 cases, as also shown in Table IV. The initial amplitude a'_{i0} and wavelength λ'_s are acquired by the same method as before. Through linear fitting, the theoretical growth rate after the dimensionless time of 0.5 in the C2E3 case is obtained, which is consistent with the experimental and numerical results. However, the model fails to predict the growth rate of the amplitude in the C2E1 case. Because of the combination of the RS with the rippled IS, the strength of the rippled IS is higher, resulting in a larger postshock Atwood number A^+ and a larger jump velocity of interface Δv than those calculated by 1D gas dynamics theory. As a result, the theoretical growth rate of the amplitude in the C2E1 case is lower than experimental and numerical values. In conclusion, the distance between cylinders influences the shock-shock interactions and consequently the amplitude growth rate. Specifically, as the distance becomes smaller, the shock-shock interactions will be more complicated, resulting in a larger amplitude growth rate.

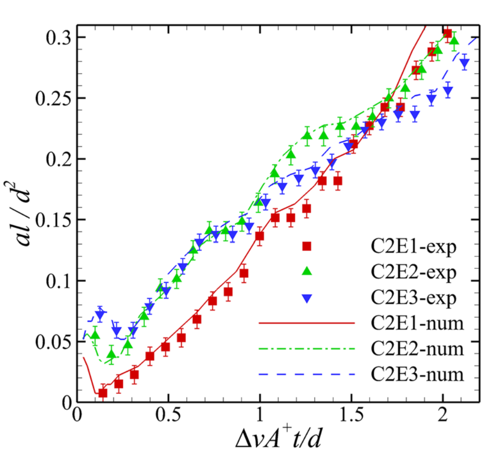

FIG. 17. Time-variations of distorted interface amplitudes for the C2E1, C2E2, and C2E3 cases.

V. CONCLUSIONS

The development of a flat air/SF_6 interface accelerated by a rippled shock wave is investigated. In experiments, the rippled shock wave is formed by diffracting a planar shock wave around solid cylinder(s), and the flat interface is generated by the soap film technique. Schlieren photography combined with high-speed imaging exhibits legible experimental pictures. Numerical simulations are also adopted to provide more information of the flow field.

First, the wave patterns after a planar shock wave diffracting around a single solid cylinder and double solid cylinders are discussed, and the differences in variations of the strength and amplitude of the rippled shock wave are illustrated. The amplitudes of rippled shock waves decrease sharply for both cases at first and then slowly in the single cylinder case but still greatly in the double cylinder case due to the great change of the Mach stem. The continuous and great reduction of the amplitude in the double cylinder case even results in a negative amplitude, i.e., a phase inversion of the shock front. Therefore, the coupling of multiple solid cylinders leads to a diverse evolution of the rippled shock wave compared with the one solid cylinder case. For the double cylinder case, the shock stability problem in the present work is analytically discussed, and the theory can only roughly predict the shock behaviors because the initial conditions and the postshock flow parameters are different. Further, the effects of the spacing between cylinders on shock stability are highlighted. The results indicate that the moment and the position from the cylinder when the phase inversion of the shock front occurs are both delayed as the spacing between cylinders increases.

Then, the evolution of a flat air/SF_6 interface impacted by a rippled shock wave diffracting around a single solid cylinder is investigated. To obtain the amplitude growth rate theoretically, the initial amplitude and wavelength of the deformed interface induced by the rippled shock are determined by extensive numerical simulations. It turns out that by using the equivalent initial amplitude of the interface, instead of the amplitude of the rippled shock front, fairly good agreement on linear growth rates is achieved among experiments, computations, and theoretical predictions. Moreover, the effects of the planar shock Mach number on the amplitude growth rate are numerically investigated. The results show that the initial interface amplitude imprinted by the rippled shock wave and the linear growth rate are both proportional to the shock Mach number. Meanwhile, the startup process is extended as the enhancement of the shock intensity, and this conclusion was also reached in classical RM instability problems.

Finally, the effects of coupling of multiple solid cylinders and spacing between cylinders on the evolution of the deformed interface are emphasized. Schlieren images show that chevron-like interfaces are generated, and the shape of connection between chevron-like interfaces is dependent on the spacing of solid cylinders. Further studies show that the coupling of cylinders will induce the shock-shock interactions which will exert effects on interface movements.

In addition, the results indicate that a smaller spacing of cylinders will promote the amplitude growth, which may be ascribed to the induction by complicated shock-shock interactions. A modified impulsive theory is proposed to predict the growth rate of the interface amplitude considering different numbers of cylinders and different distances between cylinders. The revised model is effective except the case with a small spacing of cylinders because of more complicated shock-shock interactions.

This work only discusses the effects of the cylinder number and spacing between cylinders on the interface evolution. For a single cylinder, the previous work[16] pointed out that the distance from the cylinder to the interface and the size of the cylinder will affect the results. For multiple cylinders, the effects of the two factors on the interface evolution will be investigated in near future. Also a sinusoidal rippled shock wave will be produced by adjusting the distance between the cylinder and interface, and the interaction of sinusoidal rippled shock waves with flat interfaces will be investigated to provide a universal theory for the relationship of the amplitude and wavelength between the rippled shock wave and deformed interface. Besides, the interaction of rippled shock waves with initial perturbed interfaces will also be conducted.

ACKNOWLEDGMENTS

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